Cryptanalyzing a bit-level image encryption algorithm based on chaotic maps

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Abstract

Recently, a bit-level image encryption algorithm based on chaotic maps was proposed [1], in which a diffusion-permutation structure with only one round is adopted. In this paper, we performed a comprehensive cryptanalysis on the BCIEA. Theoretical analysis and experimental simulations verify that BCIEA cannot resist a chosen ciphertext attack method.

Keywords: image encryption, cryptanalysis, bit-level, chaos.

1. Introduction

The remainder of the paper is organized as follows. Section 2 reviews the cryptosystem proposed in [2]. Then, Sec. **??** presents a comprehensive cryptanalysis on BCIEA. Section **??** verifies the effectiveness of the attack method. The last section concludes the paper.

2. The cryptosystem under study

This section fist presents the basic principle of bitplane decomposition, and then introduces BCIEA briefly.

2.1. Bitplane decomposition and composition

2.2. Description of BCIEA

The block diagram of BCIEA is shown in Fig. . As can be seen, the main components of BCIEA are introduced briefly as follows:

- *The chaotic map and the Secret Key*:

  In BCIEA, a chaotic map is used twice to generate several keystreams for encryption. The piece-wise linear chaotic map (PWLCM) is mathematically modeled by

$$x_i = F(x_{i-1}, \eta) = \begin{cases} x_{i-1}/\eta, & 0 < x_{i-1} < \eta \\ (x_{i-1} - \eta)/(0.5 - \eta), & \eta \le x_{i-1} < 0.5 \\ F(1 - x_{i-1}, \eta), & 0.5 \le x_{i-1} < 1 \end{cases} \tag{1}$$

  where x is the state variable, $x_0 \in (0, 1)$ is the initial value and $\eta \in (0, 0.5)$ is the control parameter, respectively. In BCIEA, two different groups of initial values and control parameters $(x_0, \eta_1, y_0, \eta_2)$ are served as the secret keys.

- *Initialization*: For the sake of generality, the encrypted object is an 8-bit grayscale image of size $\boldsymbol{P}$ of size $M \times N$ (height × width). The initialization operation consists of two parts: one is converting the plain image into its bit-level form, and the other is generating the keystreams by the chaotic map for the first time.

  - By BBD, convert the plain image $\boldsymbol{P}$ of size $M\times N$ (height × width) into two 1D sequences $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ of length $4MN$. For convenience, let L be $4MN$.
  - With the initial parameter η_1 and the initial value x_0, get a sequence $\boldsymbol{X}_1$ by iterating Eq. (1) for $N_0 + MN$ times and discard the former N_0 values, given as

$$X_1(i) = \mathrm{mod}(floor(x(i)) \times 10^{14}, 256)$$

    where $i = 1 \sim MN$, $floor$ is the function that takes an integer, and $X_1(i) \in \mathbb{Z}_{256}$. Then, two sequences $\boldsymbol{b}_1, \boldsymbol{b}_2$ of length L are obtained from $\boldsymbol{X_1}$ by the method of binary bitplane decomposition. Here, $\boldsymbol{b}_1, \boldsymbol{b}_2 \in \mathbb{Z}_2$.

- *Stage 1. diffusion*:

  Step 1. Firstly calculate the sum of $\boldsymbol{A}_2$, represented as

$$sum_1 = \sum_{i=1}^{L} A_2(i) \tag{2}$$

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where $L = 4MN$. Then run a cyclic right shift operation on $\boldsymbol{A}_1$ with sum_1 bits to obtain $\boldsymbol{A}_{11}$.

Step 2. Get $\boldsymbol{B}_1$ by performing the bitwise XOR operations between $\boldsymbol{A}_{11}$, $\boldsymbol{A}_2$ and $\boldsymbol{b}_1$, given as

$$\begin{cases} B_1(1) = A_{11}(1) \oplus A_{11}(L) \oplus A_2(1) \oplus b_1(1) \\ B_1(i) = A_{11}(i) \oplus A_{11}(i-1) \oplus A_2(i) \oplus b_1(i) \end{cases} \tag{3}$$

where $i = 2 \sim L$.

Step 3. Like Step 1, firstly compute the sum values sum_2 of $\boldsymbol{B}_1$ as

$$sum_2 = \sum_{i=1}^{L} B_1(i) \tag{4}$$

Then , get $\boldsymbol{A}_{22}$ cyclic right shifting $\boldsymbol{A}_2$ with sum_2 bits.

Step 4. Similar to Step 2, obtain $\boldsymbol{B}_2$ from $\boldsymbol{A}_{22}$, $\boldsymbol{B}_1$ and $\boldsymbol{b}_2$, modeled by

$$\begin{cases} B_2(1) = A_{22}(1) \oplus A_{22}(L) \oplus B_1(1) \oplus b_2(1) \\ B_2(i) = A_{22}(i) \oplus A_{22}(i-1) \oplus B_1(i) \oplus b_2(i) \end{cases} \tag{5}$$

where $i = 2 \sim L$.

- *Stage 2. Confusion*:

  Step 1. Firstly compute the sum values of $\boldsymbol{B}_1$ and $\boldsymbol{B}_1$ by

  $$sum = \sum_{i=1}^{L} (B_1(i) + B_2(i)) \tag{6}$$

  Further update the initial value of the chaotic map by

  $$s_0 = \mathrm{mod}(y_0 + sum/L, 1) \tag{7}$$

  Then, with the initial parameter η_2 and the initial value s_0, using Eq. (1) again, generate a chaotic sequences of length $2L$, and then process it into two integer sequences $\boldsymbol{Y}$ and $\boldsymbol{Z}$ both of length L. Here, $\boldsymbol{Y}, \boldsymbol{Z} \in [1, L]$.

  Step 2. Use $\boldsymbol{Y}$ and $\boldsymbol{Z}$ as coordinate indexes, and then get $\boldsymbol{C}_1$ and $\boldsymbol{C}_2$ by confusing the positions of the elements in $\boldsymbol{B}_2$ and $\boldsymbol{B}_1$ respectively. Exactly, given as

  $$\begin{cases} C_1(i) = B_2(Y(i)) \\ C_2(i) = B_1(Z(i)) \end{cases} \tag{8}$$

  where $i = 1 \sim L$.

  Finally, get the cipher image $\boldsymbol{C}$ from $\boldsymbol{C}_1$ and $\boldsymbol{C}_2$ by the method of BPC.

As stated in [1], the decryption process is the inverse of the encryption process, but the specific steps are not described. For more details, please refer to [1].

3. On the decryption of BCIEA

3.1. Problems with the decryption

As everyone knows, ensuring the decryption is available is a basic condition for good cryptosystems. In addition, decryption is an integral part of cryptanalysis. For this reason, we discuss the decryption part firstly.

- *Initialization*: As with encryption, two initialization operations are performed:
  - Get $\boldsymbol{C}_1$ and $\boldsymbol{C}_2$ from the cipher image $\boldsymbol{C}$ by BPD.
  - Generate the keystreams $\boldsymbol{b}_1, \boldsymbol{b}_2$ by the secret keys (η_1, x_0).

- *Stage 1. Anti-confusion*:

  Step 1. Firstly compute the sum values of $\boldsymbol{C}_1$ and $\boldsymbol{C}_2$ by

  $$sum = \sum_{i=1}^{L} (C_1(i) + C_2(i)) \tag{9}$$

  Note that the sum of $\boldsymbol{B}_1$ and $\boldsymbol{B}_2$ is equal to the sum of $\boldsymbol{C}_1$ and $\boldsymbol{C}_2$ because the confusion only changes the positions of the elements. Then, with the same secret keys (η_2, y_0), update the initial value y_0 to s_0 and further obtain the two coordinate indexes sequences $\boldsymbol{Y}$ and $\boldsymbol{Z}$.

  Step 2. Restore $\boldsymbol{B}_1$ and $\boldsymbol{B}_2$ from $\boldsymbol{C}_2$ and $\boldsymbol{C}_1$ by the anti-confuse, given by

  $$\begin{cases} B_1(Z(i)) = C_2(i) \\ B_2(Y(i)) = C_1(i) \end{cases} \tag{10}$$

  where $i = 1 \sim L$.

- *Stage 2. Anti-diffusion*:

  Step 1. Corresponding to Step 4 of the confusion, recover $\boldsymbol{A}_{22}$ from $\boldsymbol{B}_2$, $\boldsymbol{B}_1$ and $\boldsymbol{b}_2$, modeled by

  $$\begin{cases} A_{22}(1) \oplus A_{22}(L) = B_2(1) \oplus B_1(1) \oplus b_2(1) \\ A_{22}(i) \oplus A_{22}(i-1) = B_2(i) \oplus B_1(i) \oplus b_2(i) \end{cases} \tag{11}$$

  where $i = 2 \sim L$.

  Step 2. Obtain sum_2 by Eq. (4), and then get $\boldsymbol{A}_2$ by doing cyclic left shift operation on $\boldsymbol{A}_{22}$ with sum_2 bits.

  Step 3. Recover $\boldsymbol{A}_{11}$ by performing the bitwise XOR operations between $\boldsymbol{B}_1$, $\boldsymbol{A}_2$ and $\boldsymbol{b}_1$, given as

  $$\begin{cases} A_{11}(1) \oplus A_{11}(L) = B_1(1) \oplus A_2(1) \oplus b_1(1) \\ A_{11}(i) \oplus A_{11}(i-1) B_1(i) \oplus A_2(i) \oplus b_1(i) \end{cases} \tag{12}$$

  where $i = 2 \sim L$.

Step 4. Obtain sum_1 by Eq. (2), and then get $\boldsymbol{A}_1$ by doing cyclic left shift operation on $\boldsymbol{A}_{11}$ with sum_1 bits.

Finally, get the plain image $\boldsymbol{A}$ from $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ by the method of BPC.

Observing Eqs. (11) and (12), the decryption results are the bitwise XOR values of two elements, instead of the values of an element. In fact, this paradigm will lead to solutions that are not unique. To this end, the following proof is given from the perspective of algebraic analysis.

Take Eq. (11) as an example, let $x(i) = A_{22}(1)$, $\alpha(i) = B_2(i) \oplus B_1(i) \oplus b_2(i)$. Thus, Eq. (11) becomes

$$\begin{cases} x(1) \oplus x(L) = \alpha(1) \\ x(i) \oplus x(i-1) = \alpha(i) \end{cases} \tag{13}$$

where $i = 2 \sim L$, $\boldsymbol{x}, \boldsymbol{\alpha} \in \mathbb{Z}_2$. Here, $\boldsymbol{\alpha}$ is known, and $\boldsymbol{x}$ is to be determined. One can conclude that $\boldsymbol{x}$ has two solutions regardless of the value of $\boldsymbol{\alpha}$ due to the symmetry of two unknowns in this equation.

For example, when $L = 4$ and $\boldsymbol{\alpha}$="1010" is known. Then, both $\boldsymbol{x}$="1100" and $\boldsymbol{x}$="0011" are solutions of Eq. (13).

Obviously, this violates the condition that the encryption and decryption functions required by the cryptosystem are single shot.

3.2. A slight correction on the decryption

To ensure correct decryption, and refer to other similar algorithm designs [3], a reasonable correction is to replace $A_{11}(L), A_{22}(L)$ with $A_{11}(0), A_{22}(0)$ respectively, and $A_{11}(0), A_{22}(0) \in \mathbb{Z}_2$ also act as secret keys. Thus, without changing the idea of the algorithm, the equations of encryption are corrected to:

$$B_1(i) = A_{11}(i) \oplus A_{11}(i-1) \oplus A_2(i) \oplus b_1(i) \tag{14}$$

and

$$B_2(i) = A_{22}(i) \oplus A_{22}(i-1) \oplus B_1(i) \oplus b_2(i) \tag{15}$$

where $i = 1 \sim L$. Accordingly, the equations of decryption Eqs. (11) and (12) become

$$A_{22}(i) = A_{22}(i-1) \oplus B_2(i) \oplus B_1(i) \oplus b_2(i) \tag{16}$$

and

$$A_{11}(i) = A_{11}(i-1) \oplus B_1(i) \oplus A_2(i) \oplus b_1(i) \tag{17}$$

where $i = 1 \sim L$. This ensures the correctness of the decryption.

4. Cryptanalysis

For the adversary, the goal is to restore the acquired cipher images to the original plain images. A common way is to get its real or equivalent key. In BCIEA, the secret keys are $(x_0, \eta_1, y_0, \eta_2, A_{11}(0), A_{22}(0))$ considering the correction for the decryption given in Sec. 3.2. In fact, the difficulty of finding $(x_0, \eta_1, y_0, \eta_2)$ is quite large, even impossible, because the preproces operation as shown in Equation 1 is used. To do this, we turn our goal to get their equivalent keys, exactly $(\boldsymbol{b}_1, \boldsymbol{b}_2, A_{11}(0), A_{22}(0))$ for the diffusion and $(\boldsymbol{Y}, \boldsymbol{Z})$ for the confusion respectively.

4.1. Analysis on the diffusion part

The generation process of the two diffusion sequences $\boldsymbol{b}_1, \boldsymbol{b}_2$ are independent of the plain images. Also, $A_{11}(0)$ and $A_{22}(0)$ are not related to plain images. Thus, $(\boldsymbol{b}_1, \boldsymbol{b}_2, A_{11}(0), A_{22}(0))$ can be considered as a permanent equivalent key of the diffusion under a given secret key.

Based on chosen ciphertext attack, one can choose the all-zero cipher image $\boldsymbol{C}^{(0)}$ and get its corresponding plain image $\boldsymbol{A}^{(0)}$. Then, one gets $\boldsymbol{B}_1^{(0)} = 0$ and $\boldsymbol{B}_2^{(0)} = 0$ because the confusion does not change the element value. Also, $\boldsymbol{A}_1^{(0)}$ and $\boldsymbol{A}_2^{(0)}$ are also known according to the basic assumption of the cipher is public. At this point, the original algorithm is actually a diffusion-only process. Eq. (16) becomes

$$\begin{aligned} A_{22}^{(0)}(1) &= A_{22}(0) \oplus B_2^{(0)}(1) \oplus B_1^{(0)}(1) \oplus b_2(1) \\ &= A_{22}(0) \oplus b_2(1) \end{aligned} \tag{18}$$

$$\begin{aligned} A_{22}^{(0)}(i) &= A_{22}^{(0)}(i-1) \oplus B_2^{(0)}(i) \oplus B_1^{(0)}(i) \oplus b_2(i) \\ &= A_{22}^{(0)}(i-1) \oplus b_2(i) \end{aligned} \tag{19}$$

where $i = 2 \sim L$. Here, one has $sum_2 = 0$ because $\boldsymbol{B}_1^{(0)} = 0$. Thus, $\boldsymbol{A}_2^{(0)} = \boldsymbol{A}_{22}^{(0)}$. Moreover, $A_{22}(0)$ is fixed for a given key. Let $\hat{b}_2(1) = A_{22}(0) \oplus b_2(1)$, Eq. (18) evolves to

$$\hat{b}_2(1) = A_{22}(0) \oplus b_2(1) = A_{22}^{(0)}(1)$$

And Eq. (19) evolves to

$$b_2(i) = A_{22}^{(0)}(i) \oplus A_{22}^{(0)}(i-1)$$

where $i = 2 \sim L$. Thus, one can determine the result of $A_{22}(0) \oplus b_2(1)$, namely $\hat{b}_2(1)$, and $b_2(i)$ for $i = 2 \sim L$. Note that $A_{22}(0)$ and $b_2(1)$ always appear in pairs, so there is no need to ask for specific $A_{22}(0)$ and $b_2(1)$. Here, let $\hat{\boldsymbol{b}}_2$ be a sequence identical to $\boldsymbol{b}_2$ except that only the first element is different, exactly is $\hat{b}_2(1) = A_{22}(0) \oplus b_2(1)$.

Secondly, one gets $\boldsymbol{A}_{11}^{(0)}$ from $\boldsymbol{A}_1^{(0)}$ by doing cyclic right shift with sum_1 bits. Here, sum_1 associated with $\boldsymbol{A}_2^{(0)}$ and

$\boldsymbol{A}_1^{(0)}$ are both known, thus $\boldsymbol{A}_{11}^{(0)}$ is determined. Further, similarly following Eq. (17), Let $\hat{b}_1(1) = A_{11}(0) \oplus b_1(1)$, one has

$$\hat{b}_1(1) = A_{11}^{(0)}(1) \oplus A_2^{(0)}(1)$$

When $i = 2 \sim L$, one has

$$b_1(i) = A_{11}^{(0)}(i) \oplus A_{11}^{(0)}(i-1)$$

Similarly, let $\hat{\boldsymbol{b}}_1$ be a sequence identical to $\boldsymbol{b}_1$ except that only the first element is different, exactly is $\hat{b}_1(1) = A_{11}(0) \oplus b_1(1)$.

Consequently, another equivalent key $\hat{\boldsymbol{b}}_1, \hat{\boldsymbol{b}}_2$ are determined. They can be used to eliminate the diffusion part for any given images. This also laid the foundation for the entire cryptanalysis. As a result, these two sequences are different for different plaintext or ciphertext.

4.2. Analysis on the confusion part

Once the diffusion can be eliminated, BCIEA degenerates into a confusion-only/permutation-only algorithm. Referring to existing research, many permutation-only encryption algorithms are unable to defend against known plaintext and selective plaintext attacks. However, the objects deciphered in these studies are static replacement processes that are unrelated to plaintext. These analytical methods may not be directly applicable.

Unlike the diffusion part, the confusion keystreams $\boldsymbol{Y}, \boldsymbol{Z}$ are dynamic for the different plain or cipher images because of Eqs. (9) and (7). Thus, there is no global/permanent equivalent key for the confusion. Nevertheless, by observing Eqs. (9) and (7), one can see that This dynamic attribute also has certain rules. More, according to the given ciphertext, the sum value can be obtained, thereby providing a basis for selecting the ciphertext. This is also an important reason for choosing a ciphertext attack.

The purpose of the attacker is to restore the original information of the target ciphertext. By Eq. (9), the value of *sum* for the target cipher image $\boldsymbol{C}$ can be calculated. Thus, The diffusion sequences $\boldsymbol{Y}, \boldsymbol{Z}$ is identical under the premise of selecting ciphertexts having the same sum value. Then, the following analysis will select a ciphertext attack to select a special ciphertext with the same sum value as the given ciphertext.

For a simple example, given $L = 8$ and $\boldsymbol{Y} = (8, 4, 5, 7, 2, 1, 6, 3)$. One can choose the cipher images as

$$\begin{cases} \boldsymbol{C}_1^{(1)} = (0, 1, 0, 1, 0, 1, 0, 1) \\ \boldsymbol{C}_1^{(2)} = (0, 0, 1, 1, 0, 0, 1, 1) \\ \boldsymbol{C}_1^{(3)} = (0, 0, 0, 0, 1, 1, 1, 1) \end{cases} \tag{20}$$

Simultaneously, construct the other part $\boldsymbol{C}_2^{(1)}, \boldsymbol{C}_2^{(2)}, \boldsymbol{C}_2^{(3)}$ so that the sum of the chosen cipher images are all *sum*. Thus, one of the forms is

$$\begin{cases} \boldsymbol{C}_2^{(1)} = (1, 1, 1, 1, 1, 0, 0, 0) \\ \boldsymbol{C}_2^{(2)} = (1, 1, 1, 1, 1, 0, 0, 0) \\ \boldsymbol{C}_2^{(3)} = (1, 1, 1, 1, 1, 0, 0, 0) \end{cases} \tag{21}$$

Correspondingly, one gets $\boldsymbol{B}_2$ by performing anti-confusion on $\boldsymbol{C}_1$, given by

$$\begin{cases} \boldsymbol{B}_2^{(1)} = (0, 1, 0, 1, 0, 1, 0, 1) \\ \boldsymbol{B}_2^{(2)} = (0, 0, 1, 1, 0, 0, 1, 1) \\ \boldsymbol{B}_2^{(3)} = (0, 0, 0, 0, 1, 1, 1, 1) \end{cases} \tag{22}$$

In the confusion, the elements of $\boldsymbol{B}_1$ are only replaced by $\boldsymbol{C}_2$, and the elements of $\boldsymbol{B}_2$ are only replaced by $\boldsymbol{C}_1$. Thus, one can get the confusion sequence $\boldsymbol{Y}$ by comparing all the elements in Eq. (20) and Eq. (22).

Similarly, the cipher images are constructed by exchanging $\boldsymbol{C}_1$ in Eq. (20) and $\boldsymbol{C}_2$ in Eq. (21), thus the above method can be used to find the confusion sequence $\boldsymbol{Z}$.

4.3. The proposed attack method

Based on the above, the diffusion and the confusion of BCIEA cannot resist a chosen ciphertext attack method. Thus, for a given cipher image $\hat{\boldsymbol{C}}$, the specific steps for breaking BCIEA by chosen ciphertext attack is given as follows:

- Step 1. As Sec. 4.1, get the equivalent diffusion key $\hat{\boldsymbol{b}}_1, \hat{\boldsymbol{b}}_2$ with the all-zero chosen cipher image and the corresponding plain image.
- Step 2. Construct the $2\lceil \log_2(L) \rceil$ chosen cipher images with the same *sum* value as $\hat{\boldsymbol{C}}$ by the method presented in Sec. 4.2, and then use the decryption machine of BCIEA to get their corresponding plain images.
- Step 3. Obtain their intermedia ciphertexts $\boldsymbol{B}_1$ and $\boldsymbol{B}_2$ from these plain images by using $\hat{\boldsymbol{b}}_1, \hat{\boldsymbol{b}}_2$, and then achieve the two confusion sequences $\boldsymbol{Y}$ and $\boldsymbol{Z}$ by the method given in Sec. 4.2.
- Step 4. Recover the original image $\hat{\boldsymbol{A}}$ of a given cipher image $\hat{\boldsymbol{C}}$ with the equivalent keys $\hat{\boldsymbol{b}}_1, \hat{\boldsymbol{b}}_2$ and $\boldsymbol{Y}$ and $\boldsymbol{Z}$.

Note that the chosen cipher images constructed in Step 2 may change for a given different cipher image. Nevertheless, the attack method is still available and effective. Therefore, BCIEA cannot resist chosen ciphertext attack.

For the data complexity of the attack method, the number of cipher images required is $1 + 2\lceil \log_2(L) \rceil$. Moreover, in the terms of computational complexity, using the decryption machine is $\boldsymbol{O}(1 + 2\lceil \log_2(L) \rceil)$, determining the two equivalent keys $(\hat{\boldsymbol{b}}_1, \hat{\boldsymbol{b}}_2)$ and $(\boldsymbol{Y}, \boldsymbol{Z})$ are both $\boldsymbol{O}(2L)$, and restoring the original image is $\boldsymbol{O}(8L)$.

5. Experimental results and discussion

5.1. Experimental setup

To verify the effectiveness of the proposed attack method, some experiments were carried out. The experimental environment is Matlab R2016a based on a personal computer. Same as those in [1], experimental image is an 8-bit grayscale image “Lenna” of size 256×256. Moreover, some other images with different size and type selecting from MISC image database. considering that the secret keys of the cryptosystem are not fixed and unknown for attackers, some different keys are selected for experimental simulations.

5.2. Breaking BCIEA for the same image as [1]

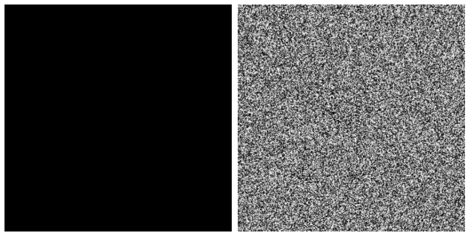
As [1], an 8-bit grayscale image “Lenna” of size 256×256 was adopted as the typical experimental object. Accordingly, the image is also served as an example for the breaking experiments. For the adversary, the task is to recover the original version of the given cipher image $\hat{\boldsymbol{C}}$, which is shown in Fig. 1(b).

(a) $\hat{\boldsymbol{A}}$ (b) $\hat{\boldsymbol{C}}$

Figure 1: The all-zero cipher image and the corresponding plain image.

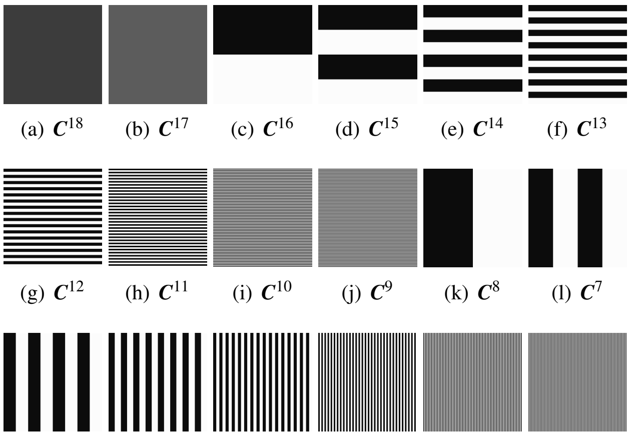
Firstly, by Step 1 of Sec. 4.3, the all-zero chosen cipher image $\boldsymbol{C}^0$ and the corresponding plain image $\boldsymbol{A}^0$ of size 256×256 are shown in Fig. 2(a)-(b) respectively. Hence, one gets the diffusion keystreams $\hat{\boldsymbol{b}}_1, \hat{\boldsymbol{b}}_2$. Secondly, by calculating $\hat{\boldsymbol{C}}$, one has $sum = 32765$. Thus, with the method of Step 2 in Sec. 4.3, two groups of the chosen cipher images with this same value are given in Fig. 3(a)-(r) and Fig. 4(a)-(r) respectively. For the sake of simplicity, the illustrations of A are not given because they look similar to Fig. 2(b), exactly are noise-like images. Thirdly, one can get the confusion sequences $\boldsymbol{Y}$ and $\boldsymbol{Z}$ by Step 3 of Sec. 4.3. Finally, by Step 4 of Sec. 4.3, the original image is restored as shown in Fig. 1(a).

For the complexity of the attack method, the number of the chosen cipher images required is 37, and the total breaking time is about 3.27s. As can be seen, the attack method is both effective and efficient for breaking BCIEA.



(a) $\boldsymbol{C}^0$ (b) $\boldsymbol{A}^0$

Figure 2: The all-zero cipher image and the corresponding plain image.

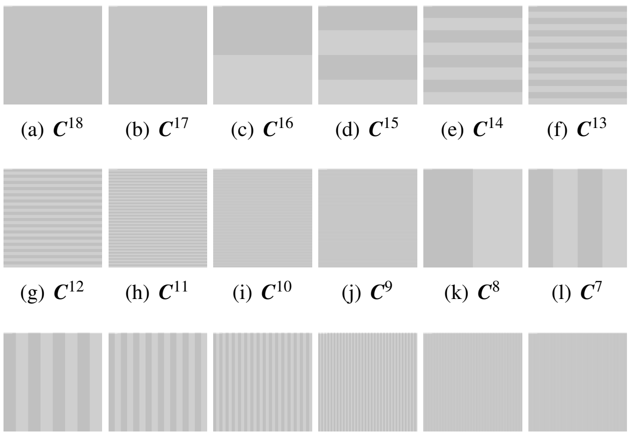


(a) $\boldsymbol{C}^{18}$ (b) $\boldsymbol{C}^{17}$ (c) $\boldsymbol{C}^{16}$ (d) $\boldsymbol{C}^{15}$ (e) $\boldsymbol{C}^{14}$ (f) $\boldsymbol{C}^{13}$

(g) $\boldsymbol{C}^{12}$ (h) $\boldsymbol{C}^{11}$ (i) $\boldsymbol{C}^{10}$ (j) $\boldsymbol{C}^{9}$ (k) $\boldsymbol{C}^{8}$ (l) $\boldsymbol{C}^{7}$

(m) $\boldsymbol{C}^{6}$ (n) $\boldsymbol{C}^{5}$ (o) $\boldsymbol{C}^{4}$ (p) $\boldsymbol{C}^{3}$ (q) $\boldsymbol{C}^{2}$ (r) $\boldsymbol{C}^{1}$

Figure 3: The 18 chosen cipher images for determining $\boldsymbol{Y}$.



(a) $\boldsymbol{C}^{18}$ (b) $\boldsymbol{C}^{17}$ (c) $\boldsymbol{C}^{16}$ (d) $\boldsymbol{C}^{15}$ (e) $\boldsymbol{C}^{14}$ (f) $\boldsymbol{C}^{13}$

(g) $\boldsymbol{C}^{12}$ (h) $\boldsymbol{C}^{11}$ (i) $\boldsymbol{C}^{10}$ (j) $\boldsymbol{C}^{9}$ (k) $\boldsymbol{C}^{8}$ (l) $\boldsymbol{C}^{7}$

(m) $\boldsymbol{C}^{6}$ (n) $\boldsymbol{C}^{5}$ (o) $\boldsymbol{C}^{4}$ (p) $\boldsymbol{C}^{3}$ (q) $\boldsymbol{C}^{2}$ (r) $\boldsymbol{C}^{1}$

Figure 4: The 18 chosen cipher images for determining $\boldsymbol{Y}$.

5.3. Breaking BCIEA for other images

To further confirm the effectiveness of the attack method, some other images selected from the MISC database were also tested. Fig. 5(a)-(f) show the attacking results for the three images. Table 1 gives the results for attacking BCIEA with some different images. It can be seen that, for these images with different types and sizes, the attack method is valid, and the required time cost is low.

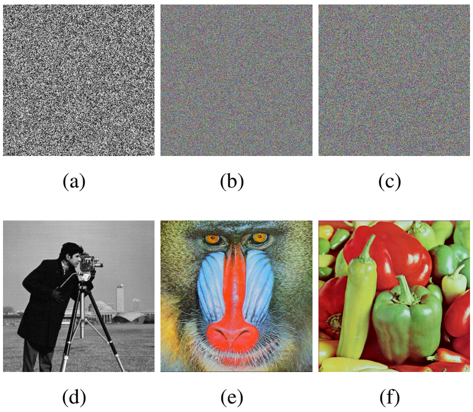


Figure 5: Experimental results for breaking BCIEA.

Table 1: Attacking results for breaking BCIEA.

Images	Encryption/Breaking times	Complexity
Lenna	0.0647s / 0.2539s	37
Cameramen	U 0.0647s / 0.2539s	41
Baboon	0.0647s / 0.2539s	37

6. Conclusions

This paper performed a comprehensive cryptanalysis on a bit-level image encryption scheme based on chaotic maps so-called BCIEA. We found that BCIEA cannot resist a chosen ciphertext attack because of its inherent security defects. Both theoretical analysis and experimental verification were provided to support that BCIEA can be broken with low computational and data complexity. This study can provide a reference for improving the security of image encryption scheme based on bit-level technique.

[4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 3, 17, 18]

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